Sourav Kumar Jena

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Computer Science and Engineering
B.Tech
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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech.	Indian Institute of Technology, Kharagpur	6.08	2025
Senior Secondary	JNV (Bundi), CBSE	76.8%	2017
Secondary	JNV (Bhadrak), CBSE	9.80	2015

EXPERIENCE

- **Freelance** *Aug 2022 – Dec 2024*
  Web & Android App Developer (Remote), Kharagpur
  – Gained experience in React.js and Android development, delivering solutions for education and e-commerce clients.
  – Developed user-friendly web and mobile apps using React.js, Node.js, MongoDB, Firebase, and native Android.
  – Built and crafted responsive UI/UX, and integrated services like payment gateways and cloud storage.
  – Collaborated with clients to gather requirements, iterate on feedback, and provide long-term support and updates.

- **Brainy Monkeys Technologies Pvt Ltd** *May 2022 – Jul. 2022*
  Web Developer Intern (Remote), Pune, Maharashtra
  – Developed responsive UI/UX using HTML, CSS, JavaScript, and jQuery for real-world client projects.
  – Worked with backend teams to integrate APIs and improve performance across browsers and devices.
  – Attended weekly technical meetings and contributed ideas for upcoming features and improvements.

PROJECTS

- **Smart Home Automation System (IoT)** *Aug. 2024 – Nov. 2024*
  Prof. Sudip Misra, IIT Kharagpur
  – Developed a smart home system using Arduino, including RFID-based access control with LCD status display.
  – Automated fan, lighting, and fire detection systems using sensors, improving energy efficiency and safety.
  – Used I2C communication protocol for seamless microcontroller integration and real-time status updates.

- **Rail Graph: Indian Railways Network Visualization** *Jan. 2024 – May 2024*
  Prof. Saptarshi Ghosh, IIT Kharagpur
  – Built a graph of Indian Railways stations from scraped Cleartrip data using Python and NetworkX.
  – Visualized routes with nodes as stations and edges as train connections, colored by traffic density.
  – Analyzed train flow patterns and plotted interactive diagrams for network load distribution.

- **Intrusion Detection Using Machine Learning** *Mar. 2022 – May 2022*
  Prof. Pabitra Mitra, IIT Kharagpur
  – Built an ML-based system to detect Web and Wireless DDoS attacks using packet analysis.
  – Used Hadoop MapReduce for large-scale traffic data processing; developed post-mortem log analysis.
  – Employed Python, Wireshark, and nmap for feature extraction and anomaly classification.

SKILLS

- **Programming Languages:** Python | Bash | SQL
- **Security & Networking Tools:** Wireshark | nmap | nslookup | arp | netstat | Burp Suite | Metasploit
- **Frameworks & Libraries:** Scapy | TensorFlow (for anomaly detection) | PyTorch
- **Databases & Cloud:** MySQL | MongoDB | AWS | GCP
- **Tools & Platforms:** Git | GitHub | Docker | Virtual Box
- **IDEs & Environments:** VS Code | Windsurf | PyCharm
- **Operating Systems:** Windows | Linux (Kali, Ubuntu)
- **Soft Skills:** Critical Thinking | Attention to Detail | Problem Solving | Communication | Collaboration

CERTIFICATIONS

- **Learn Ethical Hacking From Scratch,** Udemy *Course*
- **Website Hacking / Penetration Testing,** Udemy *Course*
- **Open Source Software Development, Linux and Git Specialization,** Coursera *Specialization*

ACHIEVEMENTS

- **Daan Scholarship,** 100% scholarship for IIT-JEE preparation *2015 – 2017*